Nutritional factors in relation to endometrial cancer: A report from a population-based case-control study in Shanghai, China

We evaluated the role of dietary nutrients in the etiology of endometrial cancer in a population-based case-control study of 1,204 newly diagnosed endometrial cancer cases and 1,212 age frequency-matched controls. Information on usual dietary habits was collected during an in-person interview using a validated, quantitative food frequency questionnaire. Logistic regression analysis was conducted to evaluate the association of nutrients with endometrial cancer risk using an energy density method (e.g., nutrient intake/1,000 kilocalories of intake). Higher energy intake was associated with increased risk, which was attributable to animal source energy and a high proportion of energy from protein and fat. Odds ratios comparing highest versus lowest quintiles of intake were elevated for intake of animal protein (Odds ratio (OR) 5 2.0, 95% confidential interval: 1.5–2.7) and fat (OR 5 1.5, 1.2–2.0), but reduced for plant sources of these nutrients (OR 5 0.7, 0.5–0.9 for protein and OR 5 0.6, 0.5–0.8 for fat). Further analysis showed that saturated and monounsaturated fat intake was associated with elevated risk, while polyunsaturated fat intake was unrelated to risk. Dietary retinol, β-carotene, vitamin C, vitamin E, fiber, and vitamin supplements were inversely associated with risk. No significant association was observed for dietary vitamin B1 or vitamin B2. Our findings suggest that associations of dietary macronutrients with endometrial cancer risk may depend on their sources, with intake of animal origin nutrients being related to higher risk and intake of plant origin nutrients related to lower risk. Dietary fiber, retinol, β-carotene, vitamin C, vitamin E, and vitamin supplementation may decrease the risk of endometrial cancer.